



ASSIGNMENT 1: PERSUASIVE TECHNOLOGY HUMAN COMPUTER INTERACTION (SECV2113-02)

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THE TECHNOLOGY:



SOLAR PANEL

a) The purpose:

The purpose of solar panels is to collect clean renewable energy in the form of sunlight and convert that light into electricity which can then be used to provide power for electrical loads that consist of energy particles called “photons”. They are used in a wide range of applications such as powering remote systems for a variety of purposes, remote sensing, telecommunications devices, and mainly for providing electricity by residential and commercial solar electric systems.

b) Feature 1:

The main features of the solar panel is to provide electricity while ensuring that you can save costs on your energy bills, and also making a positive contribution to the environment. Solar panels work by accumulating renewable energy from the sun and converting them into electricity which can later be employed to give the power to generate electricity. An average house includes an abundant roof surface for enough solar panels to generate solar electricity. This includes the supply of all the power, excess of which is sent to the power grid to be used to power the structures during the night.

Target behavior: Saving the cost of energy bills

Ability - Money

The element of simplicity or ability is money. For a person who has limited financial resources, a target behavior that costs money is not that simple. The use of electricity has become a crucial part of our daily life. For example, starting from toasters to refrigerators, microwaves, washing machines, dishwashers, electrical chimneys, and many more appliances which are simple to use and made for the convenience of day-to-day activities use electricity to function.

Motivation - Pleasure / Pain

Pleasure and pain serve as the primary motivators in the FBM. Pain and pleasure are two extremely potent motivators. The pain caused by the quantity of electricity bills that must be paid is something we can investigate. The amount of electricity used by those with numerous electronic devices will rise dramatically. Televisions and air conditioners, for example, consume enormous amounts of electricity in a single month. The bills will therefore increase in cost. Pleasure comes from the satisfied feeling after a solar panel has been installed. The bills will reduce significantly after the installation.

Prompt - Limited-time promotion

Offering a limited-time promotion to encourage people to install solar panels at their homes can make buying solar panels easier and more appealing. With discounts or special offers available for a short period, it becomes more affordable and compelling for customers to make the investment. People feel pressured by this promotion and act fast in order to take their opportunity. As a result, more people are likely to install solar panels.

b) Feature 2:

Solar energy is generated mainly with the aid of energy that is produced from the sun. A solar power panel can operate employing the solar energy that is available from the sun. Each solar panel has multiple solar cells or silicon cells which are the key building blocks. These cells absorb the energy from the sun and convert them into electricity for powering both houses and a variety of businesses. So, to make effective decisions, it is key to know the working of solar panels and how they can be useful to power your home.

Target behavior: Maximizing energy production

Ability - Emergency savior

The ability is an emergency savior. In this world nowadays, anything can happen within a split second like a world disaster. For example, when our house is blackout due to thunderstorms, solar panels can provide us with some of the electricity because solar panels can store some energy that can last for the entire day. We can think of it like a battery.

Motivation - Hope / Fear

Hope and fear have long been powerful motivators in persuasive technology. When solar panels are installed, for example, people are driven by hope since solar energy can provide us with endless energy. Fear stems from not knowing what to do in the event of a blackout throughout the entire house.

Prompt - Reminders and educational content

The trigger to encourage the behavior of maximizing energy production through solar panels could be timely reminders and educational content about the benefits of solar energy especially in emergency situations. For example, notifications reminding users about the reliability of solar panels during blackouts and educational content from the internet or books providing tips on how to optimize energy production during different weather conditions will prompt users to take actions.

d) Failed Feature: Lack of customization or adaptability to different energy needs

Aspect of solar panel technology that might fail to persuade users to execute the target behavior is the lack of customization or adaptability to different energy needs. Users may feel less motivated to invest in solar panels if they perceive that the technology does not meet their specific energy needs or preferences. Users may struggle to adapt with the technology if they find it difficult to customize or adjust the system to suit their unique requirements. Without clear options for customization or adaptability, users may not feel prompted to invest in solar panel technology.

Recommendation: Monitoring Apps

Monitoring apps can help users to determine the optimal configuration of solar panels based on their energy usage patterns. This will increase motivation by demonstrating the potential benefits of customization. The feedback mechanism from the apps allow users to monitor the operation of their customized solar panel systems and make adjustments as needed. Therefore, this app can adapt and be customized over time without being afraid of insufficient energy usage. Picture 2 below is the example of an app that can be used to monitor the energy usage from SolarEdge.



Picture 2: Monitoring app from SolarEdge